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(yapi-kredi-teknoloji) C:\Users\hamit\Desktop\repos\interview_live_code\companies\YAPI-KREDI-TEKNOLOJI>uv run scripts/evaluate.py --collection llama3_herd --limit 3
2026-06-23 11:24:37,604 INFO [src.agent.graph] Graph compiled | nodes=[ 'prepare', 'triage', 'smalltalk', 'research', 'validate', 'respond' ] | max_retries=1
2026-06-23 11:24:37,806 INFO [src.agent.nodes] [prepare] session-eval-1 | new turn
2026-06-23 11:24:37,807 INFO [src.agent.triage] Triage | question='what is the parameter count of the flagship Llama 3 model, and on roughly how ma' | has_context=False
2026-06-23 11:24:39,921 INFO [src.agent.triage] Triage result | intent=document | reply_len=0
2026-06-23 11:24:39,922 INFO [src.agent.nodes] [triage] intent=document
2026-06-23 11:24:39,923 INFO [src.agent.tools] Initializing retriever for collection='llama3 herd'
2026-06-23 11:24:40,096 DEBUG [src.retrieval.embedding] Embedder initialized: model=text-embedding-3-small
2026-06-23 11:24:40,650 DEBUG [src.retrieval.indexer] Indexer initialized: persist_directory=data\llama3_herd\chroma
2026-06-23 11:24:40,694 INFO [src.agent.nodes] [research] built agent for collection='llama3 herd' (description=True)
2026-06-23 11:24:41,684 INFO [src.agent.tools] Tool triggered [search_semantic] | Query: 'Llama 3 flagship model parameter count tokens pre-trained' | Section: None
2026-06-23 11:24:41,685 DEBUG [src.retrieval.retriever] Retrieve (dense): top_k=3 where=None query='Llama 3 flagship model parameter count tokens pre-trained'
2026-06-23 11:24:41,685 DEBUG [src.retrieval.embedding] Embedding batch: 1 texts
2026-06-23 11:24:44,618 DEBUG [src.retrieval.embedding] Received 1 embeddings (dim=1536)
2026-06-23 11:24:44,740 INFO [src.retrieval.retriever] Retrieved 3 chunk(s) for query
2026-06-23 11:24:46,380 INFO [src.agent.nodes] [research] steps=4 | tool_msgs=1
2026-06-23 11:24:46,381 INFO [src.agent.validation] Grounding check | chunks=3 | draft_len=116
2026-06-23 11:24:47,842 INFO [src.agent.validation] Grounding result | grounded=True | confidence=1.00 | unverified=0
2026-06-23 11:24:47,843 INFO [src.agent.nodes] [validate] grounded | confidence=1.00
2026-06-23 11:24:47,845 INFO [src.agent.nodes] [respond] grounded=True | answer_len=116
```

Q1: What is the parameter count of the flagship Llama 3 model, and on roughly how many tokens was it pre-trained?

EXPECTED: The flagship model has 405B parameters and was pre-trained on a corpus of approximately 15.6 trillion (15T+) multilingual tokens.

ANSWER: The flagship Llama 3 model has **405B** parameters and was pre-trained on **15.6T** text tokens [llama3_eval_p1_c1].

Grounded: True | Confidence: 100% | Citations: llama3_eval_p1_c1

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2026-06-23 11:24:47,849 INFO [src.agent.nodes] [prepare] session-eval-2 | new turn
2026-06-23 11:24:47,850 INFO [src.agent.triage] Triage | question='what attention mechanism does llama 3 adopt to improve inference scalability, an' | has_context=False
2026-06-23 11:24:49,025 INFO [src.agent.triage] Triage result | intent=document | reply_len=0
2026-06-23 11:24:49,025 INFO [src.agent.nodes] [triage] intent=document
2026-06-23 11:24:49,027 INFO [src.agent.tools] Initializing retriever for collection='llama3 herd'
2026-06-23 11:24:49,202 DEBUG [src.retrieval.embedding] Embedder initialized: model=text-embedding-3-small
2026-06-23 11:24:49,211 DEBUG [src.retrieval.indexer] Indexer initialized: persist_directory=data\llama3_herd\chroma
2026-06-23 11:24:50,370 INFO [src.agent.tools] Tool triggered [search_semantic] | Query: 'Llama 3 attention mechanism inference scalability key-value heads' | Section: None
2026-06-23 11:24:50,370 DEBUG [src.retrieval.retriever] Retrieve (dense): top_k=5 where=None query='Llama 3 attention mechanism inference scalability key-value heads'
2026-06-23 11:24:50,370 DEBUG [src.retrieval.embedding] Embedding batch: 1 texts
2026-06-23 11:24:50,575 DEBUG [src.retrieval.embedding] Received 1 embeddings (dim=1536)
2026-06-23 11:24:50,579 INFO [src.retrieval.retriever] Retrieved 5 chunk(s) for query
2026-06-23 11:24:53,104 INFO [src.agent.nodes] [research] steps=4 | tool_msgs=1
2026-06-23 11:24:53,105 INFO [src.agent.validation] Grounding check | chunks=5 | draft_len=355
2026-06-23 11:24:55,510 INFO [src.agent.validation] Grounding result | grounded=True | confidence=1.00 | unverified=0
2026-06-23 11:24:55,510 INFO [src.agent.nodes] [validate] grounded | confidence=1.00
2026-06-23 11:24:55,512 INFO [src.agent.nodes] [respond] grounded=True | answer_len=227
```

Q2: What attention mechanism does Llama 3 adopt to improve inference scalability, and how many key-value heads does it use?

EXPECTED: It uses Grouped Query Attention (GQA) with 8 key-value heads, which speeds up inference and reduces the size of the key-value cache during decoding.

ANSWER: Llama 3 adopts **Grouped Query Attention (GQA)** to improve inference scalability. The model uses **8** key-value heads. This design choice helps improve inference speed and reduces the size of key-value caches during decoding.

Grounded: True | Confidence: 100% | Citations: llama3_eval_p6_c14

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2026-06-23 11:24:55,515 INFO [src.agent.nodes] [prepare] session-eval-3 | new turn
2026-06-23 11:24:55,516 INFO [src.agent.triage] Triage | question='which preference-optimization algorithm does llama 3 use for alignment during po' | has_context=False
2026-06-23 11:24:56,591 INFO [src.agent.triage] Triage result | intent=document | reply_len=0
2026-06-23 11:24:56,592 INFO [src.agent.nodes] [triage] intent=document
2026-06-23 11:24:56,593 INFO [src.agent.tools] Initializing retriever for collection='llama3 herd'
2026-06-23 11:24:56,792 DEBUG [src.retrieval.embedding] Embedder initialized: model=text-embedding-3-small
2026-06-23 11:24:56,804 DEBUG [src.retrieval.indexer] Indexer initialized: persist_directory=data\llama3_herd\chroma
2026-06-23 11:24:58,086 INFO [src.agent.tools] Tool triggered [search_semantic] | Query: 'preference optimization algorithm alignment post-training PPO' | Section: None
2026-06-23 11:24:58,086 DEBUG [src.retrieval.retriever] Retrieve (dense): top_k=5 where=None query='preference optimization algorithm alignment post-training PPO'
2026-06-23 11:24:58,086 DEBUG [src.retrieval.embedding] Embedding batch: 1 texts
2026-06-23 11:24:59,447 DEBUG [src.retrieval.embedding] Received 1 embeddings (dim=1536)
2026-06-23 11:24:59,460 INFO [src.retrieval.retriever] Retrieved 5 chunk(s) for query
2026-06-23 11:25:03,732 INFO [src.agent.nodes] [research] steps=4 | tool_msgs=1
2026-06-23 11:25:03,734 INFO [src.agent.validation] Grounding check | chunks=5 | draft_len=644
2026-06-23 11:25:07,009 INFO [src.agent.validation] Grounding result | grounded=True | confidence=1.00 | unverified=0
2026-06-23 11:25:07,009 INFO [src.agent.nodes] [validate] grounded | confidence=1.00
2026-06-23 11:25:07,011 INFO [src.agent.nodes] [respond] grounded=True | answer_len=644
```

Q3: Which preference-optimization algorithm does Llama 3 use for alignment during post-training, and does it rely on PPO?

EXPECTED: Post-training aligns the model with supervised finetuning (SFT) followed by Direct Preference Optimization (DPO). The authors deliberately use DPO rather than on-policy RL algorithms such as PPO, finding it more stable and cheaper to run at this scale.

ANSWER: Based on the document, Llama 3 uses **Direct Preference Optimization (DPO)** for alignment during post-training. [llama3_eval_p16_c37]

Regarding PPO: No, Llama 3 does not rely on PPO. The document explicitly states that the team "explored on-policy algorithms such as PPO (Schulman et al., 2017), but found that DPO required less compute for large-scale models and performed better, especially on instruction following benchmarks like IFEval." [llama3_eval_p16_c37]

The post-training strategy involves several rounds where each round includes supervised finetuning (SFT) followed by Direct Preference Optimization (DPO). [llama3_eval_p15_c34]

Grounded: True | Confidence: 100% | Citations: llama3_eval_p16_c37, llama3_eval_p15_c34

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Thread 019ef396-efdb-7260-9ba5-8cea16df329e

Interact Trace

Run experiment

graph TD; start([__start__]) --> prepare[prepare]; prepare --> triage[triage]; triage --> research[research]; triage --> validate[validate]; research --> smalltalk[smalltalk]; smalltalk --> triage; validate --> respond[respond]; respond --> end([__end__]);

Input

View Raw

Question

Required

Input

Collection

Required

Manage Assistants

Submit

Thread 019ef396-efdb-7260-9ba5-8cea16df329e

Interact Trace

Run experiment

> question What GPU hardware, and approximately how many of them, ...

> __start__

47 seconds ago

> prepare

47 seconds ago

> triage

43 seconds ago

> research

39 seconds ago

> validate

37 seconds ago

> respond

> final_answer

Based on the document, Llama 3 405B was trained on up to 16,000 H100 GPUs [llama3_eval_p9_c19]. Each GPU had 80GB HBM3 memory and ran at 700W TDP, and they were deployed on Meta's Grand Teton AI server platform, with eight GPUs per server connected via NVLink [llama3_eval_p9_c19].

> citations [...]

> confidence 1

> is grounded true

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Deploy

Connected

Memory

Interrupts

graph TD; start([__start__]) --> prepare[prepare]; prepare --> triage[triage]; triage --> research[research]; triage --> validate[validate]; research --> smalltalk[smalltalk]; smalltalk --> triage; validate --> respond[respond]; respond --> end([__end__]);